

Yubo Wang

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Foundation-model adaptation, multimodal and structured representation learning, reliable out-of-distribution generalization

Education

Brown University, USA

Sep 2025 – May 2027 (expected)

M.S. in Computer Science. GPA 4.0/4.0

Beijing Jiaotong University, China

Sep 2021 – Jun 2025

B.Eng. in Computer Science and Technology. GPA 3.77/4.0 (top 5%), Honors Student

Preprints and Manuscripts

- **Yubo Wang**, Jingying Ma, Xinliang Zhou, Yangxuan Zhou, Jiquan Wang, Sha Zhao, Yiyuan Yang, Yi Ding, Ziyu Jia, Chenyu Liu, Cuntai Guan. ZeroMAG: Zero-Shot Multimodal Adapter Generation for Plug-and-Play EEG Foundation Models. *arXiv:2610.03546*, 2026.
- **Yubo Wang**, Xiaoyu Tong, Xinliang Zhou, Yu Zhang[†]. Do EEG Foundation Models Generalize Across Clinical Cohorts? An External Blind-Test Study of Depression Biomarkers. *In preparation for IEEE J-BHI*.

[†]Corresponding author.

Research Experience

ZeroMAG: Zero-Shot Multimodal Adapter Generation for EEG Foundation Models

Mar 2026 – Present

Stanford SPNI Lab and Nanyang Technological University, Graduate Student Researcher

Mentors: Prof. Yu Zhang, Prof. Cuntai Guan

- Proposed a zero-shot framework that equips frozen EEG foundation models with companion physiological modalities through generated plug-and-play adapters.
- Designed a structure-aware weight VAE, a modality–subject–task-conditioned diffusion transformer, and a mixture-of-experts decoder for heterogeneous adapter parameter blocks.
- Adapts to unseen subjects, datasets and modality settings without target labels or target-side optimization, keeping the backbone frozen.
- Across 3 backbones and 6 held-out datasets, improved balanced accuracy by **+7.22 pp** over EEG-only and came within **0.50 pp** of supervised multimodal adaptation.

External Blind-Test Generalization of EEG Models across Clinical Cohorts

May 2026 – Present

Stanford SPNI Lab, Visiting Student Researcher

Mentor: Prof. Yu Zhang

- Built an external blind-test framework for depression diagnosis across independent clinical cohorts, using EEG foundation models, task-specific models and clinical covariates.
- Designed domain-adversarial learning and prediction-bias correction from unlabeled local development data, with no access to external test labels.
- Used strict subject-level evaluation to quantify cohort and site shift, and how locally strong models degrade at deployment.

PhysioRuler: Structured Physiological Representations of Whole-Night Sleep

Aug 2026 – Present

Stanford University School of Medicine and Harvard Medical School (BIDMC), Research Intern

Mentors: Prof. Haoqi Sun, Prof. M. Brandon Westover

- Developing a reference-based representation of whole-night PSG organized by sleep stage, stability, physiological system and temporal scale, rather than staging accuracy alone.
- Modeling EEG, EOG, ECG and respiration with system-specific encoders and 2/10/30/120-s multiscale aggregation; a 5-state hidden semi-Markov model and CAP-informed weak labels structure stage–stability dynamics.
- Exposing interpretable axes for slow-wave and sigma activity, respiratory variability, and cortical and cardiorespiratory instability, and testing their prospective value across heterogeneous cohorts beyond clinical covariates.

Tabula 2.0: Foundation Models for Structured Scientific and Cellular Data
Research Collaborator
Mentors: Dr. Jiayuan Ding (Stanford University), Dr. Qingsong Wen (University of Oxford)

Sep 2026 – Present

- Extending the Tabula single-cell foundation model with multimodal capabilities and larger-scale pretraining, toward a foundation model for unordered, heterogeneous scientific tables.
- Exploring discretized and latent representations with predictive and generative objectives for cellular-state modeling, including action-conditioned prediction under perturbations.

Robust EEG Learning: Cross-Subject Emotion Recognition and Shortcut Audits
Institute of Automation, Chinese Academy of Sciences, Undergraduate Research Assistant
Mentor: Prof. Ziyu Jia

Nov 2024 – Jun 2025

- Audited emotion-recognition models with controlled ablations and feature visualization, identifying shortcut learning driven by cross-channel artifacts rather than physiological patterns.
- Built a variational Bayesian heterogeneous graph neural network with relationship-distribution adaptation to align inter-channel relations across subjects.
- Reached **73.5%** average accuracy and **71.85%** F1 under leave-one-subject-out evaluation across multiple datasets.

Selected Project

MyGO: Audio Foundation Model via LeJEPA

Sep – Dec 2025

- Built a high-throughput tf.data pipeline over 20,000+ AudioSet clips with on-the-fly log-mel STFT for distributed training.
- Implemented vectorized multi-view augmentations for LeJEPA and co-developed a ViT audio encoder with decoupled frequency and time positional encodings.

Industry Experience

China Academy of Railway Sciences
Software Development Engineer Intern

Jun – Aug 2024

- Built vector-retrieval and LLM-integration components for a railway knowledge assistant: data ingestion, chunking, embedding and containerized deployment.

Advanced Computer Systems Research Center, Chinese Academy of Sciences
Intern

Dec 2022 – Mar 2023

- Developed Python data-visualization tools for research workflows and implemented the Maximal Information Coefficient (MIC) in Java.

Honors and Awards

Merit Student Award (top 5%), Beijing Jiaotong University	2022, 2023, 2024
First-Class Academic Scholarship, Beijing Jiaotong University	2022, 2023, 2024
Alumni Scholarship, Beijing Jiaotong University	2022

Technical Skills

Programming	Python, C++, Java, Bash; Git, Docker, $\text{\LaTeX}$
Deep learning	PyTorch, TensorFlow, multi-GPU and distributed training, Hugging Face, Weights & Biases
Signals and data	EEG/PSG preprocessing (MNE, BIDS), NumPy, SciPy, Pandas, scikit-learn, Matplotlib